SILICYTE

Does AI self-improvement actually compound?

Pre-registered, compute-matched tests of recursive self-improvement, with the test set sealed and every result published, including nulls.

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THE PROBLEM

Self-improvement claims are graded on the tests they were selected on.

- AI systems now write the tools, prompts and code their next round runs on.
- Published library-learning gains largely disappeared once compute was matched and reuse was measured (Berlot-Attwell et al., arXiv 2410.20274 and 2504.03048).
- In our own self-improving search, a selected gain of **`+0.0095`** read **`+0.0014`** on a sealed lockbox.

FIRST STUDY: RATCHET (SILICYTE-0119, REGISTERED 4 NOV 2026)

Does a growing library of verified helpers beat its look-alikes at equal tokens?

Arm	What it rules out
Frozen after round 1	gain without growth
Same tokens, one round	dose, not compounding
Verified examples, no helpers	context, not reuse
Matched sham library	prompt length and genre
Extra samples, no library	plain test-time compute

400 sealed ARC-AGI-1 tasks (id list hashed before any run), a reuse gate, a knockout, a contamination gate, and a planted positive control that voids the run if the harness cannot see a real library. Pre-registration:

github.com/braxtonmensah/ratchet. Result pending; our registered prediction is **+2** of 400.

WHY TRUST US

We retract our own best numbers in public.

- Our earlier headline, a guardrail hook absent on **`29.8%`** of the calls it existed to check, came out **`0.0000`** on clean public machines. We said so on the homepage.
- A verifier-gated search beat flat sampling 20 to 15 of 24 hard ARC outputs; by the rule we registered first (p = 0.0625) it is a null, and we report it as one.
- Paper: degrading a design loop's scorer made its reported gain 2.4x larger (t = **`-14.60`**; DOI 10.5281/zenodo.23441786).

FOUNDER AND STAGE

Solo, technical, pre-revenue.

- Braxton Mensah, Indiana University undergraduate. Directs an AI-agent research fleet (9,700+ cluster jobs, about 7,424 GPU-hours as of 18 Oct 2026). Trained and served his own ARC Prize 2026 system; official results 4 Dec 2026.
- $0 revenue, 0 customers, unincorporated, no money raised.
- Raising a pre-seed to run the test series behind Ratchet and to turn the sealed, compute-matched protocol into a service labs use before they announce a self-improvement result.

Design and implementation are AI-assisted; the founder sets the questions, the registered pass rules and the retractions. Not peer reviewed.